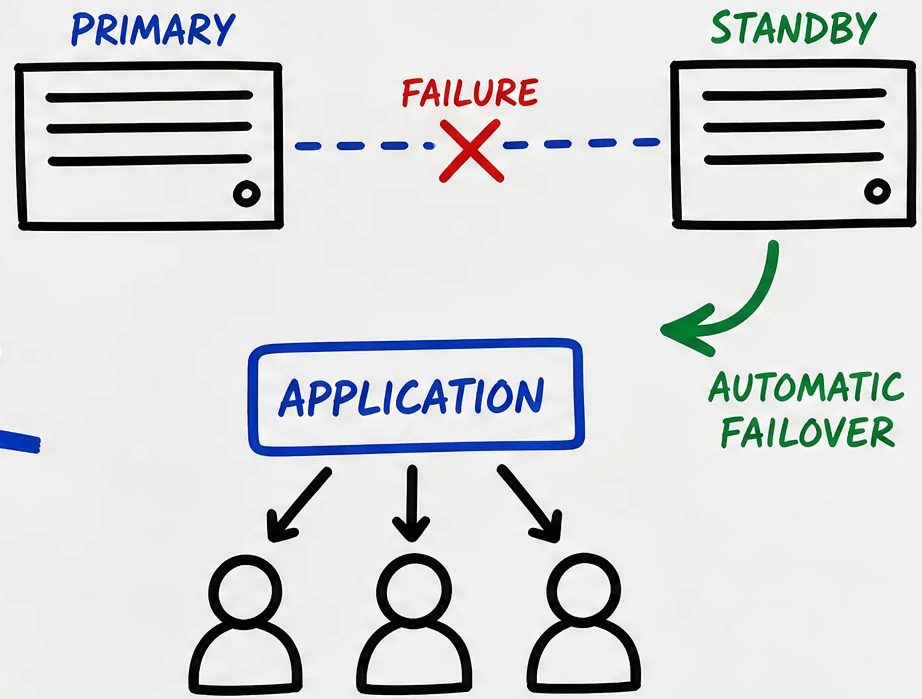
MODULE 14 – HIGH AVAILABILITY

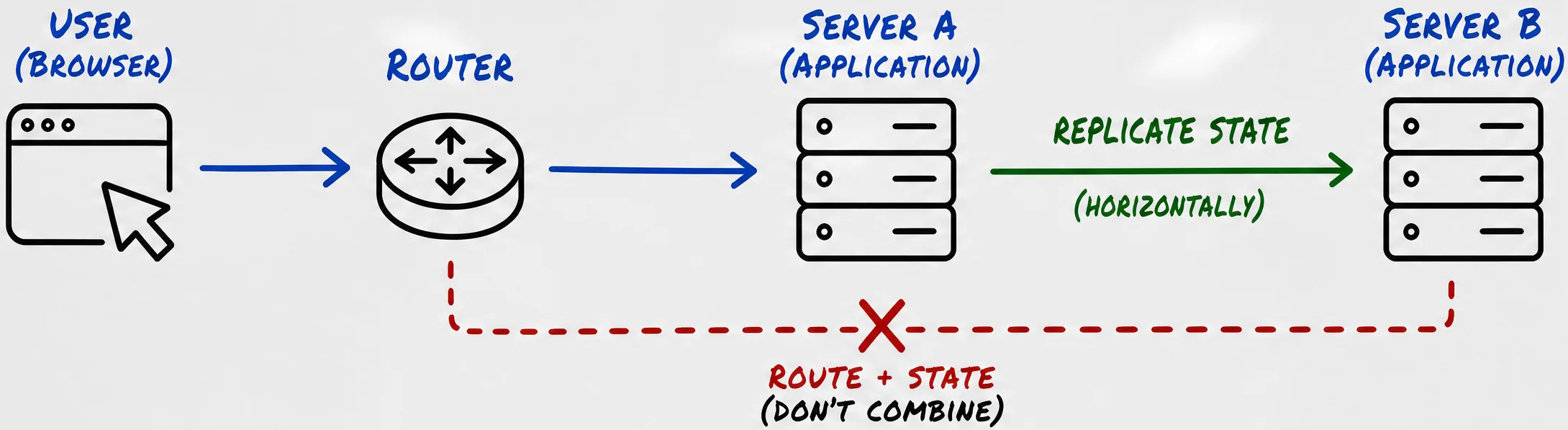
ProTech
protechtraining.com

FAILOVER IS A TESTED GUARANTEE.



AVAILABILITY IS A DELIBERATE SYSTEM PROPERTY, NOT A DEFAULT SERVER SETTING.

Route Traffic and Replicate State Separately.

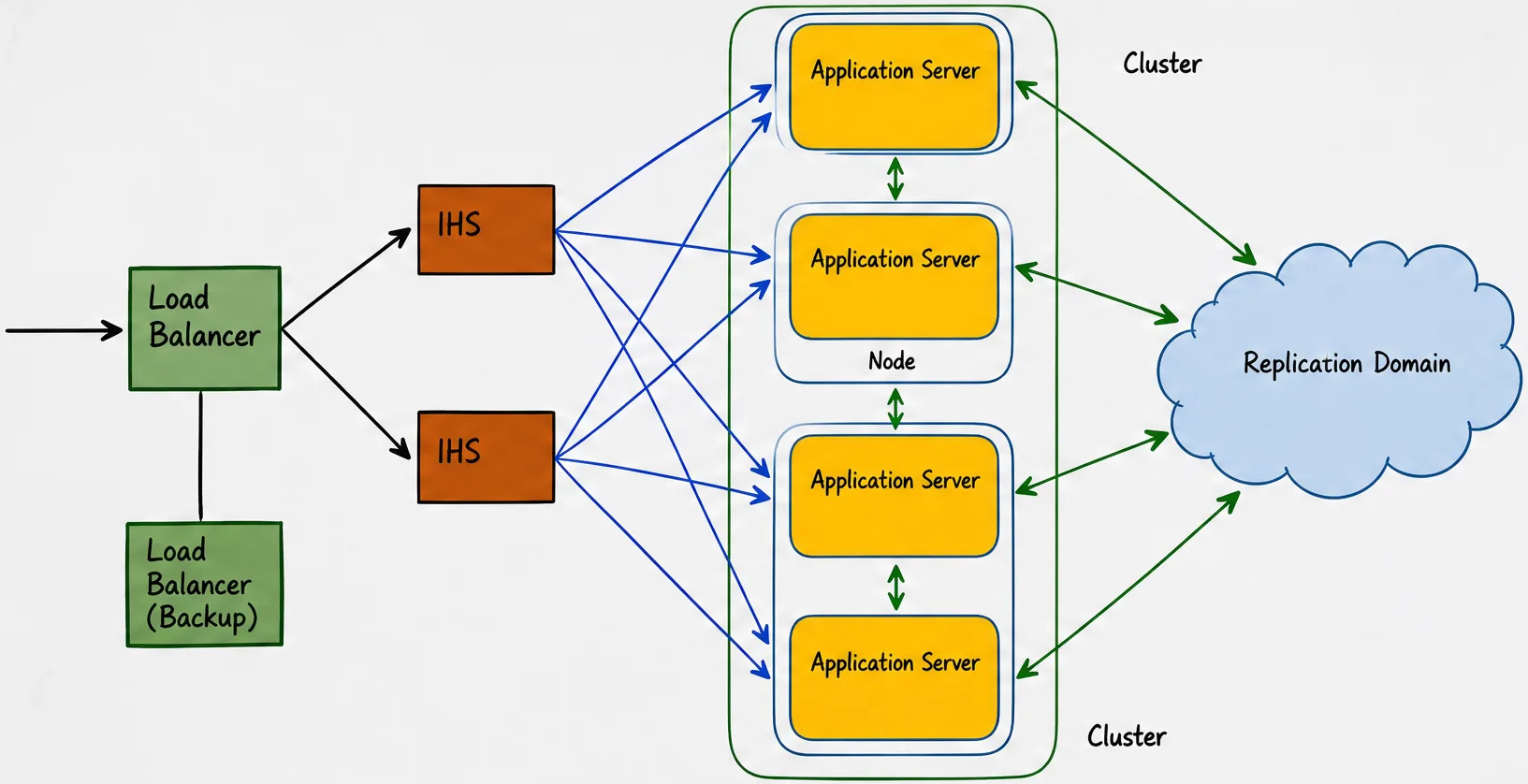


Route is NOT State.



Routing sends the user to the application;
Replication ensures the application remembers the user.

Read The Topology Before Changing It.



→ = Request Path

↔ = Replicated State

Route and State are Separate.

⚠ Traffic failover without state failover **is only half available.**

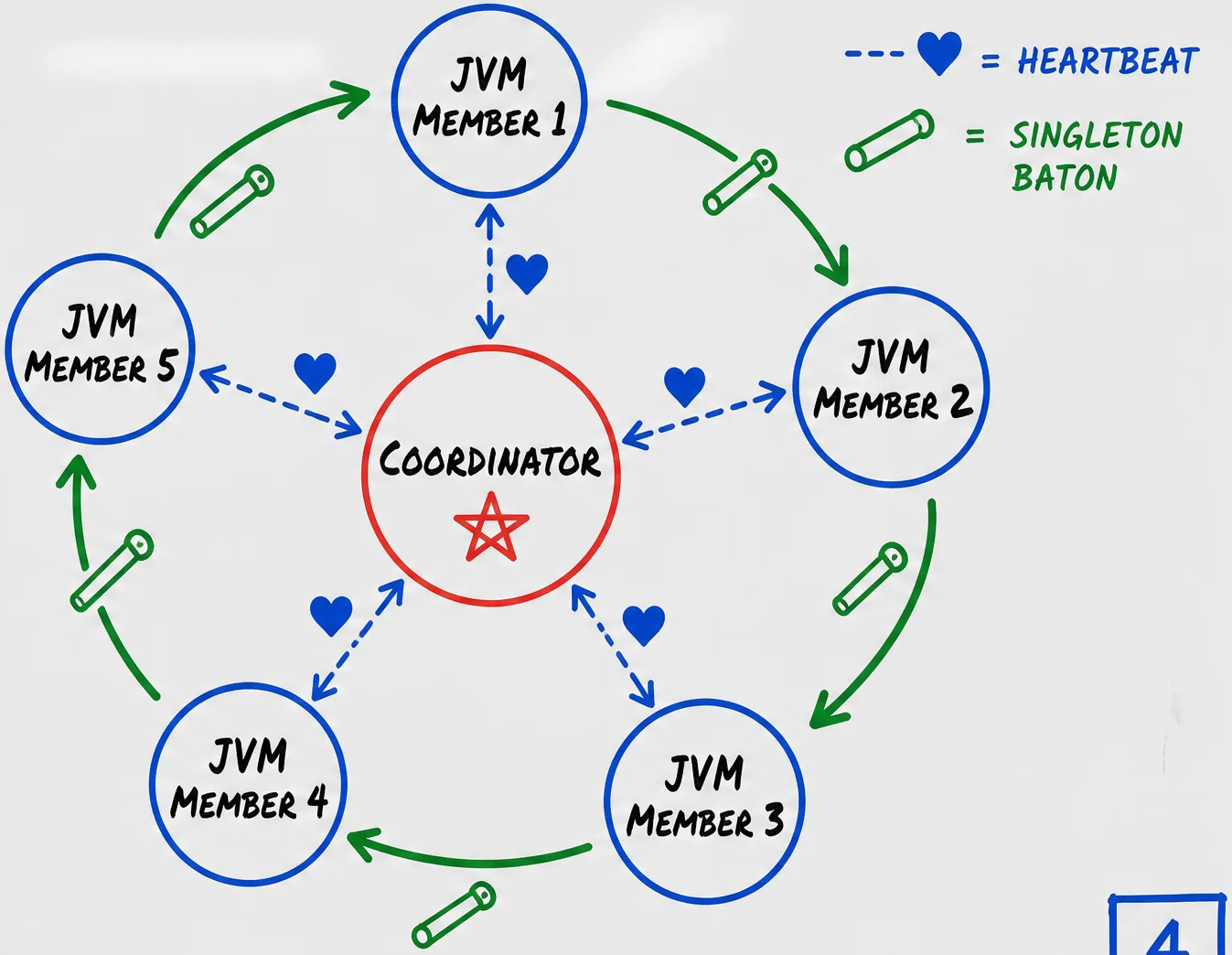
Core Groups Run the HA Control Plane.

Membership

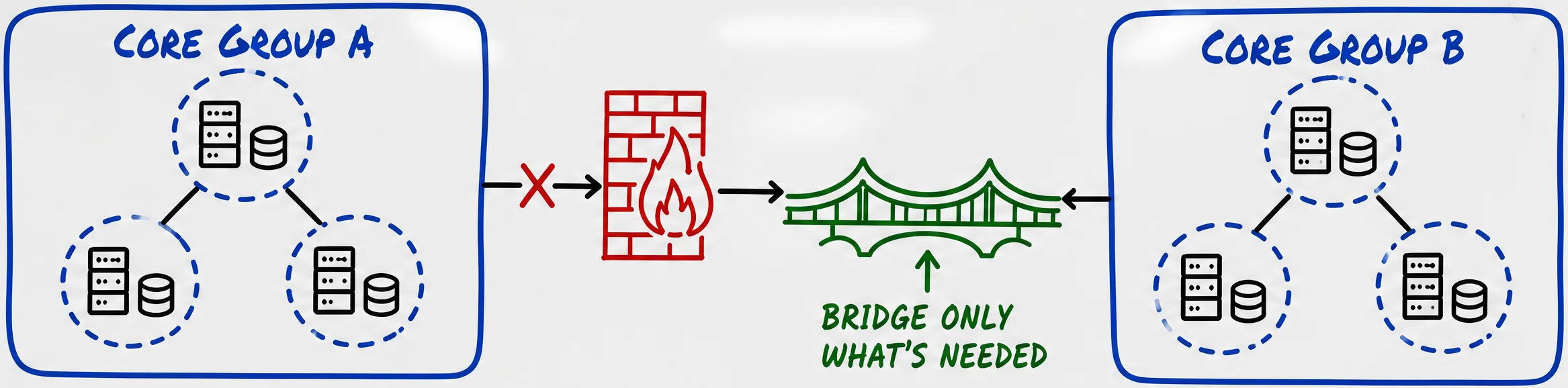
enables Takeover.



Failover starts with accurate membership.



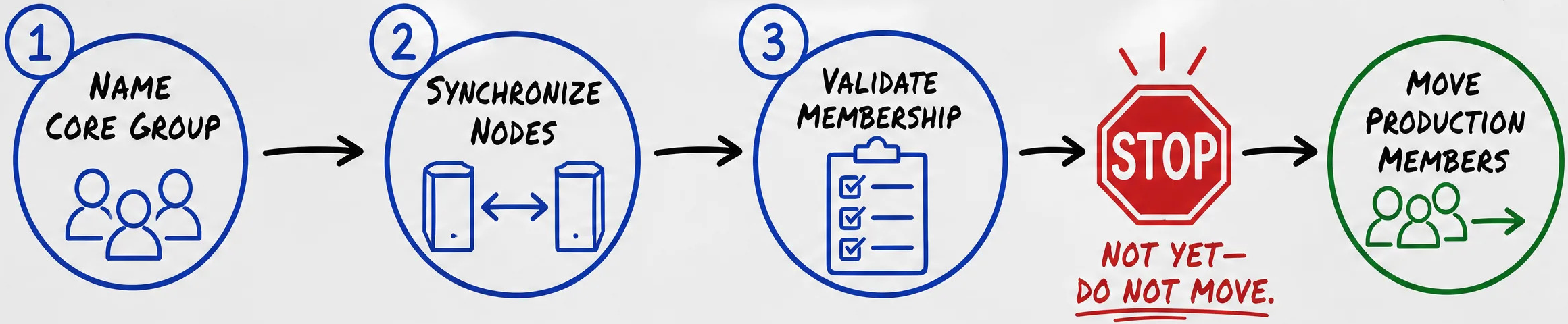
Design Core Groups Around Reachability.



Partition with Purpose.

A cluster belongs to **ONE** core group.

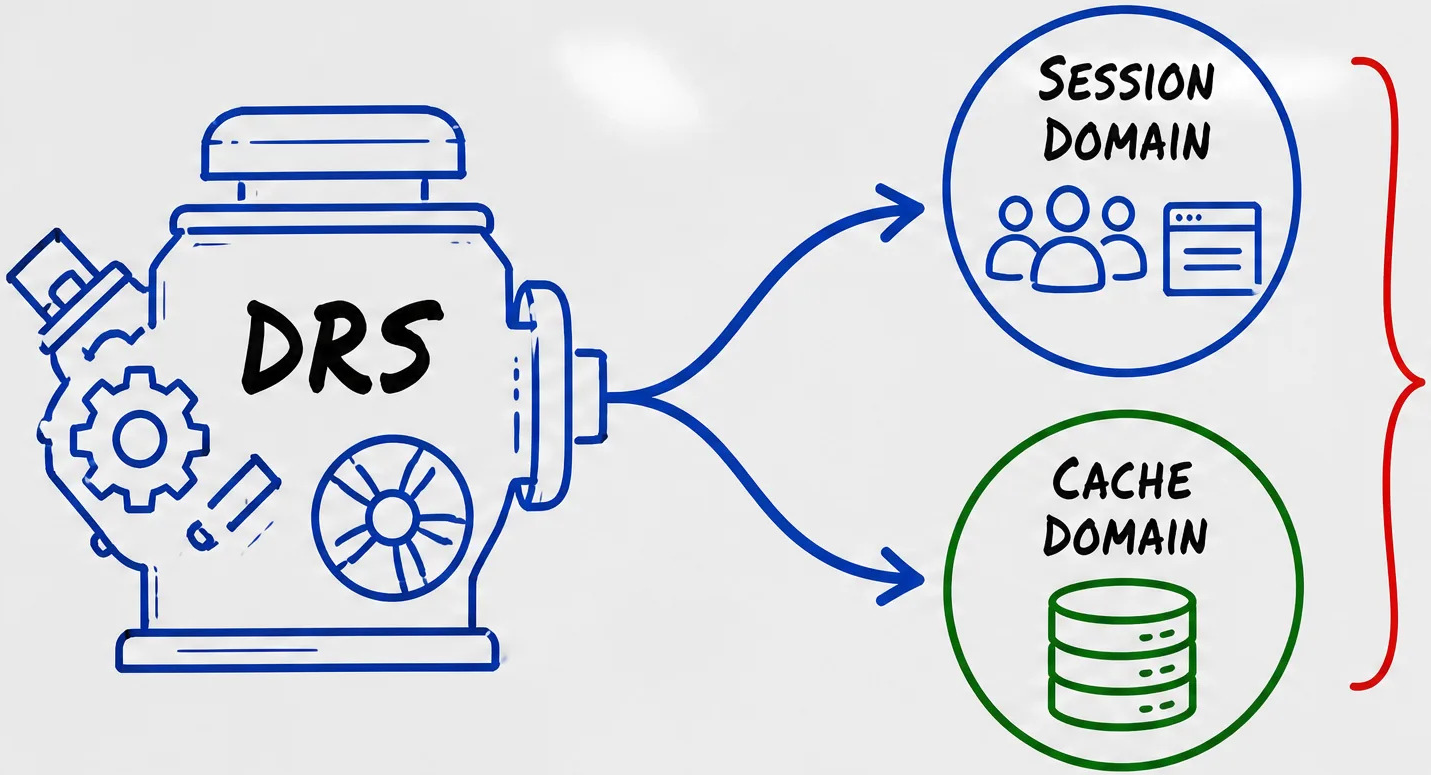
Create, Synchronize, Then Move.



Create is **not** complete.

A saved **Core Group** is only a container until its **Members** and **Policy** are verified.

DRS: One Engine, Distinct Consumers.



SHARED:

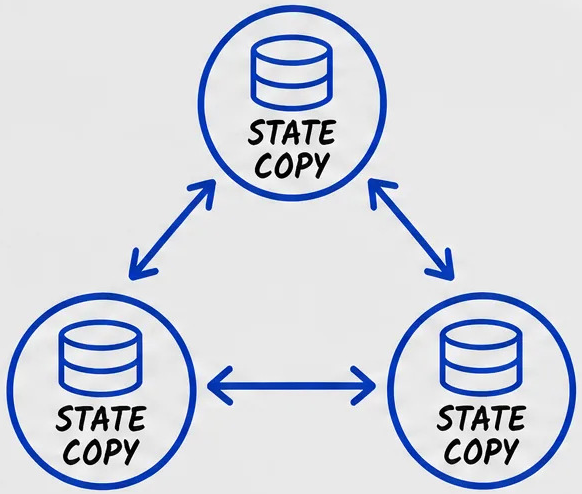
- HTTP-SESSION
- STATEFUL-EJB EXCEPTION

Separate consumers deliberately.

A replication domain is a **contract**, not a generic **broadcast bus**.

Session Replication: Choose Topology and Consistency.

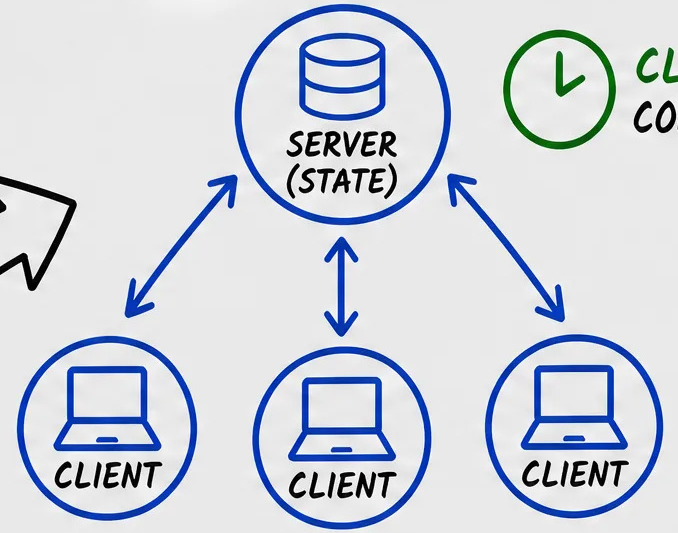
Peer-to-Peer



CLOCKS: DIVERGE

State has a cost.

Client/Server



CLOCK: CONSISTENT

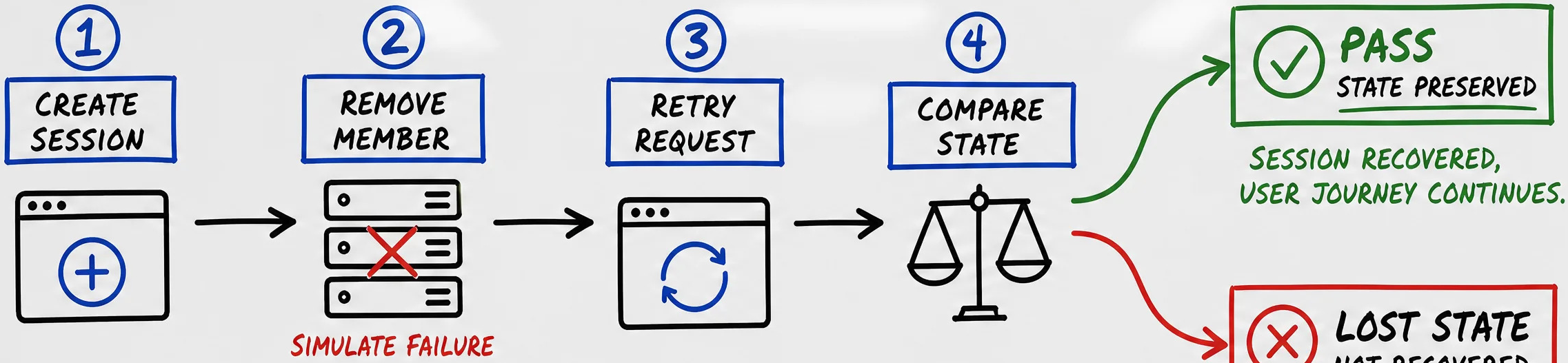


PERFORMANCE: TUNABLE



A faster write interval **narrows loss exposure** but **increases replication work.**

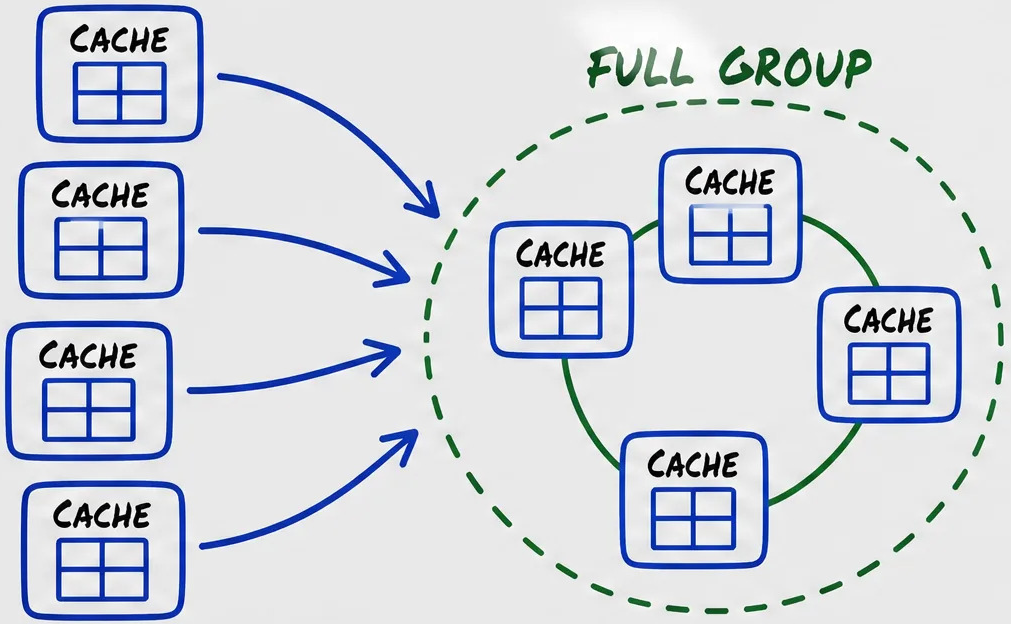
Prove Session Failover Without Guessing.



Test the user journey.

A healthy process list **does not** prove a recoverable session.

Cache Replication: Coherent Data Costs Bandwidth.



Cache Coherence is deliberate.

Invalidation Bell



Keep everyone consistent.

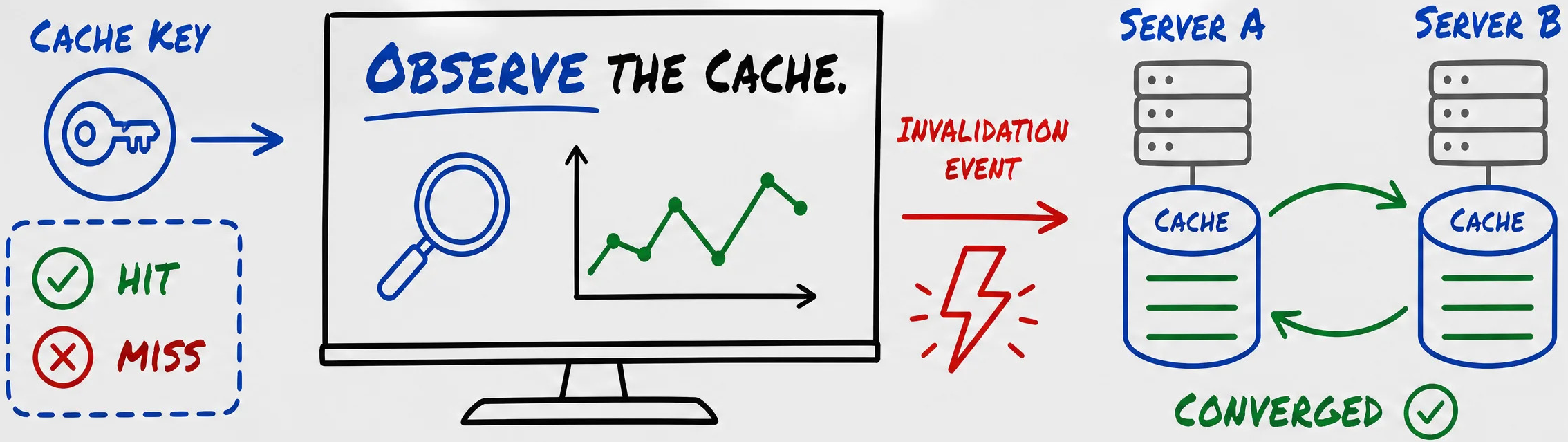
Bandwidth Meter



Coherence consumes bandwidth.

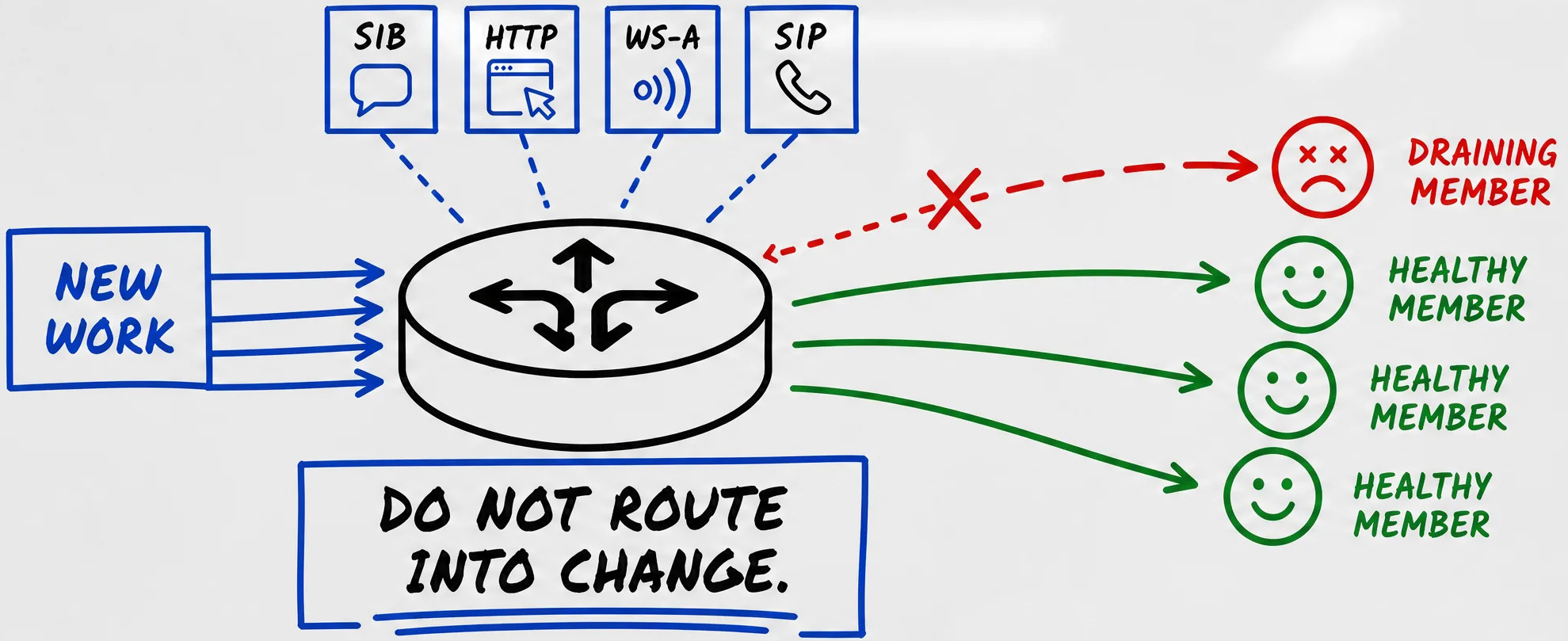
Dynamic Cache requires full-group replicas, not a single backup.

Verify Cache Replication and Invalidation.



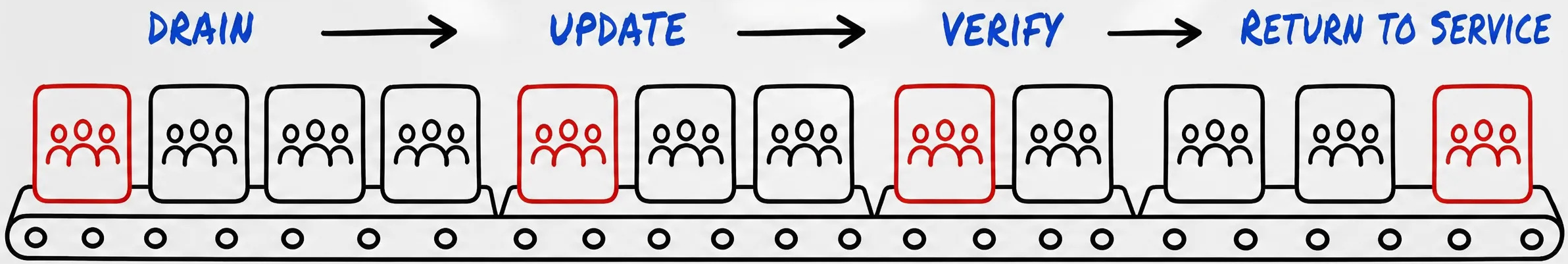
Cache Replication is correct only when updates become visible where they should.

Routing Is the Availability Switch.



☆ **AVAILABILITY** FAILS WHEN **ROUTING OUTRUNS** RUNTIME STATE.

Roll Out Updates One Safe Unit at a Time.

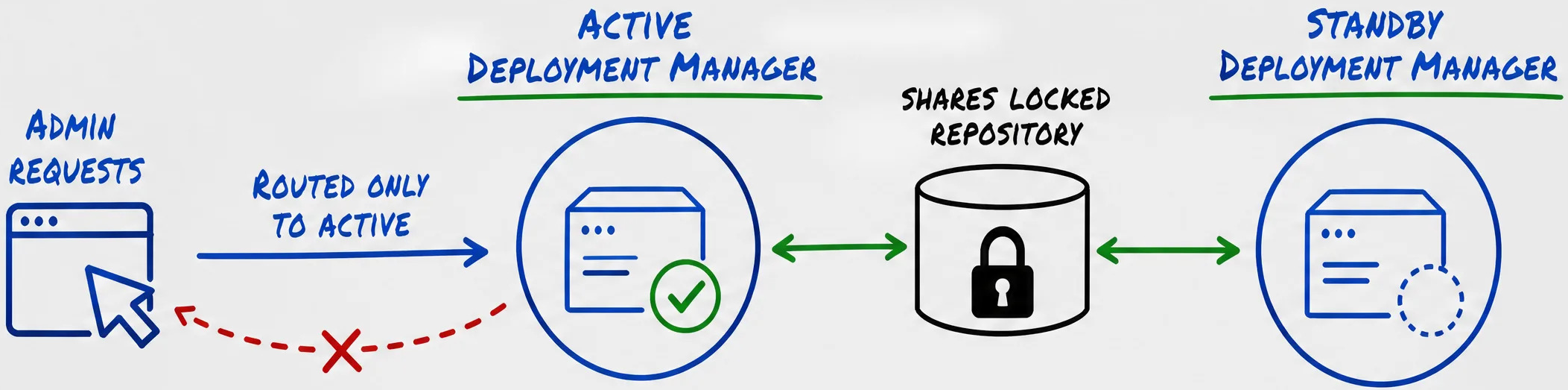


DRAIN BEFORE DEPLOY.



CONCURRENT SYNCHRONIZATION
DOES NOT GUARANTEE CONTINUOUS AVAILABILITY.

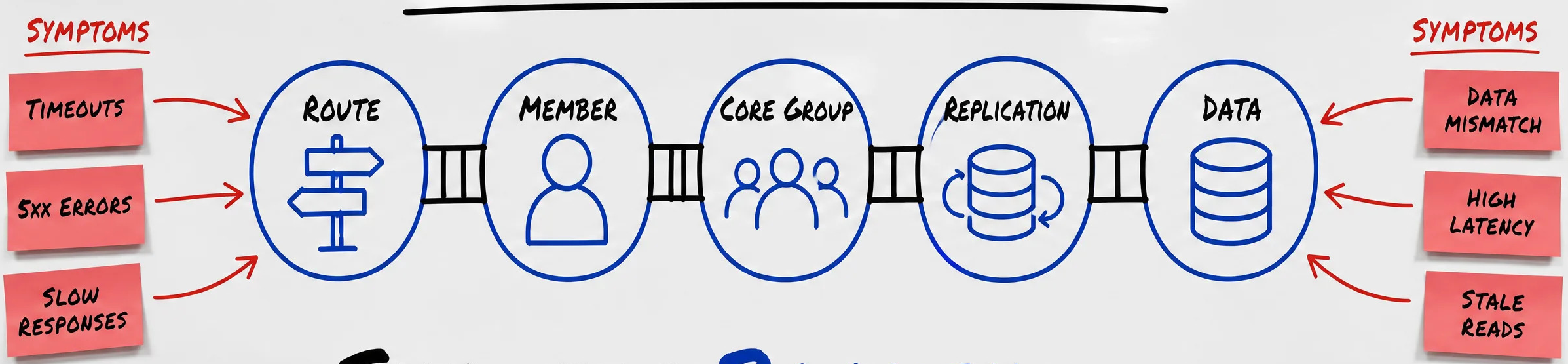
HA Deployment Manager Protects Administration.



Admin has its own **Failover**.

A **standby** deployment manager is **ready**,
not concurrently **writable**.

Troubleshoot the Broken Contract, Not the Symptom.



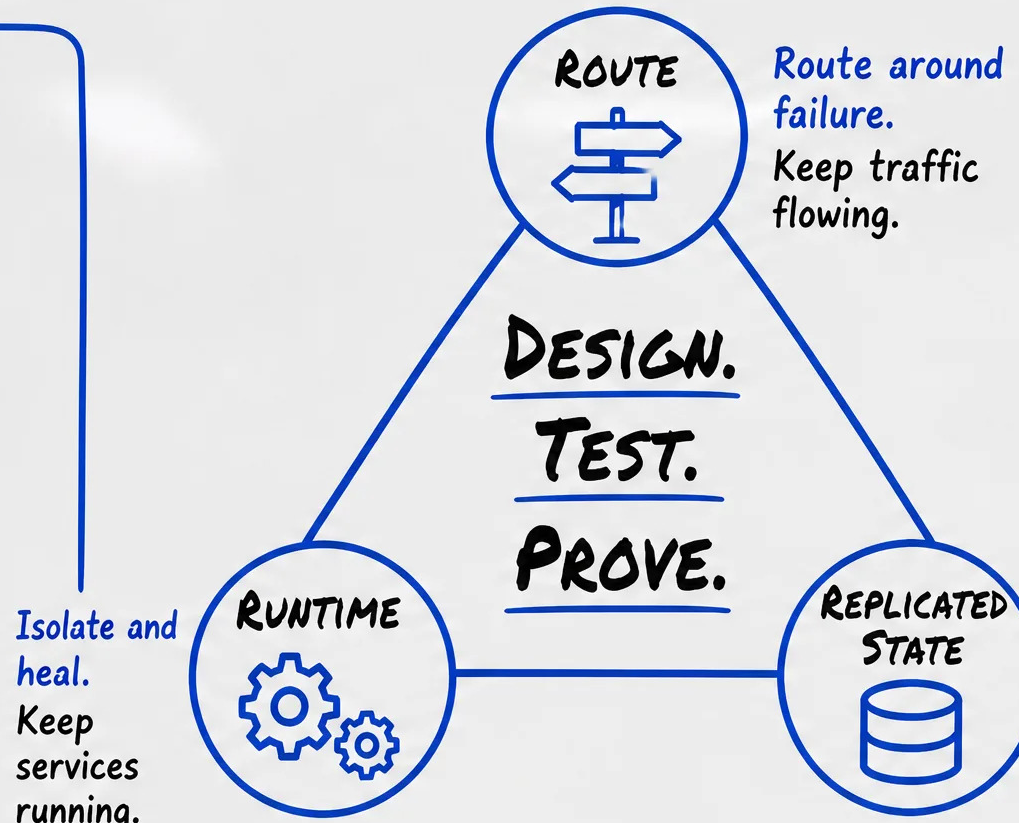
Find the Broken layer.

An outage symptom is not a root-cause category.

Recap and Lab: Survive a Server Loss

Lab Checklist

- [x] **1. Plan**
  Define scenario, success criteria, and safety guardrails
- [x] **2. Execute**
  Induce server loss and observe system behavior
- [x] **3. Verify**
  Validate continuity, data integrity, and recovery
- [x] **4. Document**
  Capture evidence, lessons learned, and improvements



Route around failure.
Keep traffic flowing.

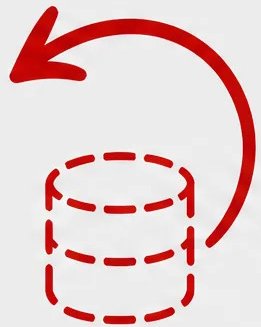
Isolate and heal.
Keep services running.

Keep data safe.
Replicate and stay consistent.

Evidence



- Logs
- Metrics
- Screenshots
- Results



Rollback
Return to last known good state if needed.



High Availability is credible only after a controlled failure test.